QUESTION: 7 WRITE A C PROGRAM FOR INSERTION OF A NEW NODE AT ANY POSITION OF A SINGLE LINKED LIST

main.c　Share　Run

```c
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node *next;
};
struct Node* insertAtPosition(struct Node *head, int value, int pos) {
    struct Node *newNode, *temp;
    int i;

    newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = value;
    if (pos == 1) {
        newNode->next = head;
        return newNode;
    }
    temp = head;
    for (i = 1; i < pos - 1 && temp != NULL; i++) {
        temp = temp->next;
    }
    if (temp == NULL) {
        printf("Invalid Position!\n");
        free(newNode);
        return head;
    }
    newNode->next = temp->next;
    temp->next = newNode;
    return head;
}
void display(struct Node *head) {
    struct Node *temp = head;
    while (temp != NULL) {
        printf("%d -> ", temp->data);
        temp = temp->next;
    }
    printf("NULL\n");
}
int main() {
    struct Node *head = NULL;
    head = insertAtPosition(head, 10, 1);
    head = insertAtPosition(head, 20, 2);
    head = insertAtPosition(head, 30, 3);
    printf("Linked List before insertion:\n");
    display(head);
    head = insertAtPosition(head, 25, 3);
    printf("Linked List after insertion at position 3:\n");
    display(head);
    return 0;
}
```

Output

```
Linked List before insertion:
10 -> 20 -> 30 -> NULL
Linked List after insertion at position 3:
10 -> 20 -> 25 -> 30 -> NULL


=== Code Execution Successful ===
```